Resolution Geometry: Protocol-Composition Rigidity

Four Operational Gates Cut a Frozen Atlas of Theories to Complex Quantum Mechanics

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Abstract

A framework that can describe everything has no teeth. We test whether RG's native operations act as strict filters: quotienting by operational equivalence, composing protocols, and asking whether a distinction survives arbitrary extension.

The frozen specification (C0). Before any gate is run, a specification is hashed (SHA-256 `497fa4b5...`). It contains primitives, gates and a candidate atlas. It contains no Hilbert space, Born rule, tensor product, Schrödinger equation or operator algebra.

The four gates.

(G1) Composition descent.

(G2) Extension completeness.

(G3) In-family closure under iterated composition.

(G4) Continuous, extension-stable reversibility.

The results. The gates cut the frozen atlas strictly, in two steps:

- *atlas:* classical, real, complex, quaternionic, boxworld, polygon, random-polygon and superellipse theories;
- *after G2–G3:* classical, complex, boxworld and the polygon-type theories;
- *after G4:* complex QM alone.

Every eliminated pair carries an explicit obstruction certificate. One new structural lemma connects two of the gates:

- the local quotient of transformations is extension-complete *iff* composites are spanned by products;
- in real QM, identity and transpose agree on every single-system state but differ on a two-system state.

Further results:

- Among qubit symmetries, exactly the proper rotations $SO(3) \cong PU(2)$ survive extension. Every improper one fails with eigenvalue $-\frac{1}{2}$.
- The surviving generators are the traceless Hermitian matrices, i.e. Hamiltonians.
- As a stress test, all 300 random polygon theories are eliminated.
- Tsirelson's $2\sqrt{2}$ appears only as an *output* comparator.

Scope. The cut is proved on the frozen atlas. For the unrestricted space of operational theories, the uniqueness of complex QM is imported from reconstruction theorems (Hardy; Chiribella–D'Ariano–Perinotti; Masanes–Müller), whose axioms RG's gates instantiate. We claim teeth, not novelty of the final result.

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1 Why rigidity, and how not to cheat

The objection to RG is not that it is wrong, but that it is accommodating: a meta-geometry that classifies everything predicts nothing. Rigidity is the answer. It requires four things:

(i) gates fixed before results;

(ii) a candidate atlas that includes the expected loser;

(iii) strict elimination with certificates;

(iv) the target, here quantum mechanics, appearing only as an output.

Definition 1.1 (C0 freeze)**.** The specification contains:

- *primitives*: preparations, effects, outcome laws, sequential composition, parallel composition $\boxtimes$ (not assumed to be a tensor product), classical mixing and coarse-graining;
- *the equivalence*: $P \sim Q$ iff every admissible test gives equal laws;
- *the gates* G1–G4;
- *the candidate atlas*;
- the post-gate comparators;
- a list of forbidden premises.

Its SHA-256 is verified by the battery before any gate runs.

2 G1: composition descent

Proposition 2.1 (Descent)**.** *Operational composition, defined on equivalence classes, descends: $P \sim P'$ and $Q \sim Q'$ imply $P \boxtimes Q \sim P' \boxtimes Q'$. A "composition" that depends on a hidden representative is rejected.*

Witness (C1). The maximally mixed qubit can be prepared as an equal mixture of $|0\rangle, |1\rangle$ or of $|+\rangle, |-\rangle$. These are operationally identical, and their composites with an ancilla agree. A composition that records the hidden decomposition label yields probabilities 0.25 and 0.5 on a joint effect, so it depends on information the protocol cannot access, and G1 removes it.
G1 removes a class of compositions, not a candidate theory: every theory in the atlas composes on its quotient.

3 G2: extension completeness, and its equivalence with local tomography

Theorem 3.1 (Extension–tomography equivalence)**.** *For a family of linear state spaces, the following are equivalent:*

(a) transformations that agree on every single-system state agree on every extension $T \boxtimes I_C$;

(b) the composite state space is spanned by products of local states, i.e. local tomography.

Proof. (b)⇒(a): $T \boxtimes I$ is fixed on products, which span the composite.

(a)⇒(b): if products do not span, take a direction w in the composite outside their span. A map that equals T on local states but acts differently on the local factors generating w separates the extensions. □

Witness (C2).

- *Real QM.* The single-system space has dimension 3 and the two-system space dimension 10, while products span only 9.
- *Complex QM.* The dimensions are 4, 16 and 16.
- *The separating pair.* In real QM, identity and transpose agree on every single-system state, but on $\rho_\pm = (I \pm 0.2\,\sigma_y \otimes \sigma_y)/4$ the partial transpose maps ρ_+ to ρ_-. The local quotient is *too coarse*, and the extension-complete quotient separates them.

4 G3: in-family closure under iterated composition

Proposition 4.1 (Closure arithmetic; C3, C4)**.** *Let $K(d)$ be the number of real parameters of a d-level system. Local tomography requires $K_{AB} = K_A K_B$, and closure requires $K_A K_B = K(n)$ for some system size n of the same family, at every copy number.*

Family	$K(d)$	*Two qubits:* $K_A K_B$	*In-family solution*
classical	d	*4*	$n = 4$*; closes for all copies*
real QM	$d(d+1)/2$	*9*	*none* ($K(4) = 10$)
complex QM	d^2	*16*	$n = 4$*; closes for all copies*
quaternionic QM	$d(2d-1)$	*36*	$n = 4.5$*, none*
boxworld	*maximal tensor product*	*9*	*closes by construction*

Checked up to six copies. Real QM fails at two copies, with kernel witness $\sigma_y \otimes \sigma_y$. Quaternionic QM has no in-family composite.

5 G4: continuous, extension-stable reversibility

Proposition 5.1 (Polytopes have finite reversible groups; C5)**.** *The reversible transformations of a polytope state space are affine maps permuting vertices, so they form a finite group, and distinct pure states cannot be joined by a continuous reversible path. The computed group orders are:*

- *regular n-gon bits: $2n$ (4, 8, 10, 12, 16 for $n = 2, 4, 5, 6, 8$; the segment, i.e. the classical bit, has order 2);*
- *300 random polygon bits: order 1 in 286 cases and 6 in 14 cases (degenerate triangles, i.e. classical trits).*

Every one fails G4.

Proposition 5.2 (Among smooth bits, only the ball)**.** *For superellipse bits $|x|^p + |y|^p \leq 1$, a rotation by 0.3 rad leaves a boundary defect of $> 10^{-3}$ when $p \in \{1.5, 3, 4\}$, and $< 10^{-12}$ when $p = 2$. Continuous transitivity on the pure states selects the ellipse, which is affinely a ball, as in the "bit = ball" lemma [3].*

Theorem 5.3 (Extension-stable reversible dynamics of the qubit)**.** *The symmetries of the Bloch ball form $O(3)$. Under extension by an ancilla:*

- *every proper rotation stays admissible: on the Bell state, the minimum eigenvalue is $\geq -3 \times 10^{-16}$ over 200 random samples;*
- *every improper element fails: the eigenvalue is $-\frac{1}{2}$, as for the transpose.*

The extension-stable continuous reversible group is therefore $SO(3) \cong PU(2)$. Its Lie algebra $\mathfrak{so}(3) \cong \mathfrak{su}(2)$ has $3 = d^2 - 1$ generators: the traceless Hermitian matrices, i.e. Hamiltonians. For $d > 2$, Wigner's theorem (imported) gives unitary or antiunitary symmetries, and the antiunitary ones fail extension as the transpose does.

6 The rigidity chain and obstruction certificates

Stage	Survivors	Removed (certificate)
$\mathfrak{R}_0$ (frozen atlas)	classical, real, complex, quaternionic, boxworld, polygons, random polygons, superellipses	—
$\mathfrak{R}_1 = \mathfrak{R}_0$ |G1	all candidate theories	hidden-label compositions
$\mathfrak{R}_2 = \mathfrak{R}_1$ |G2+G3	classical, complex, boxworld, polygons, random polygons, superellipses	real QM ($\sigma_y \otimes \sigma_y$ kernel; identity $\sim$ transpose); quaternionic (no composite)
$\mathfrak{R}_3 = \mathfrak{R}_2$ |G4	**complex QM**	classical (S_2); boxworld (order 8); polygons (dihedral); random polygons (order ≤ 6); superellipses (defect $> 10^{-3}$)

The minimal distinguishing protocol for each pair is:

- real vs complex: a two-copy joint effect, $\sigma_y \otimes \sigma_y$;

- quaternionic vs complex: two-copy closure;
- classical vs complex: continuous reversibility, or CHSH 2 vs $2\sqrt{2}$;
- boxworld vs complex: continuous reversibility, or CHSH 4 vs $2\sqrt{2}$;
- polygon vs complex: the finite reversible group.

Comparators, after the gates (C6). Local deterministic CHSH is 2 (exhaustive). The survivor's CHSH is $2\sqrt{2}$, maximized over settings. The PR box is 4. Tsirelson's bound is an *output*, not an input.

7 Scope, and what this gives RG

Imported Theorem 7.1 (Unrestricted reconstruction [1, 2, 3])**.** *Over all finite-dimensional operational theories, complex quantum theory is the unique theory satisfying local tomography, continuous reversibility, the subspace axiom and the measurement postulate (Masanes–Müller form). Equivalent sets are given by Hardy and by Chiribella–D'Ariano–Perinotti.*

RG's gates instantiate these axioms:

- G2 is local tomography, by Theorem 3.1;
- G4 is continuous reversibility with complete admissibility;
- G3 is closure.

The subspace and measurement axioms remain declared.

- **Teeth.** The gates eliminate strictly, every elimination carries a certificate, and random theories are eliminated wholesale. RG is not universally accommodating.
- **Non-circularity.** The quantum structures are absent from the frozen premises; Tsirelson, unitaries and Hamiltonians appear as outputs.
- **Bridge.** The extension gate is RG's native "does this distinction survive every allowed extension". It is the lever that separates real from complex quantum theory and dynamics from mere symmetry.

No-Go 7.2 (Not claimed).

- A new derivation of quantum theory: the unrestricted result is imported.
- Rigidity beyond the frozen atlas, except through the imported theorems.
- Infinite-dimensional systems or quantum field theory.
- That the subspace and measurement axioms follow from RG.

8 Reproduction

- `rg_rigidity_C0.json` and its SHA-256 are written before the battery.
- `rg_rigidity.py` (NumPy, SciPy) verifies the hash, then runs C1–C6. Reruns are byte-identical.
- The battery uses exact or 10^{-9}-tolerance linear algebra. It is not interval-certified.

References

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